

VPrivCal Expert Review

Interface-aligned screening workbook for VPrivCal-Q10 and VPrivCal-Probe

Short workshop version for professional researchers

Decision / research question

Are the three-option trigger questions, expert-verified exposure labels, Probe interface, and held-out acceptance evaluation sufficiently clear, low-burden, and testable?

Method at a glance

Element	Current design
VPrivCal-Q10	Ten direct policy questions; Q1-Q6 share one three-option trigger scale.
VPrivCal-Probe	Three egocentric scenes spanning private, public, and semi-public contexts. After a required Hint walkthrough, participants mark areas without naming categories, then review each present internal category through neutral linked-content labels and optional evidence highlights.
Calibration target	Provisional pilot target: approximately 5-7 minutes total for Q10 plus Probe. Time the required Hint walkthrough separately and confirm the total empirically.
Effectiveness test	Generic versus Q10-only versus full VPrivCal under the same expert-verified cue set and standardized reminder; participants report cue awareness, immediate acceptance, and preferred binary decision.
Expert output	Review the three trigger labels, exposure annotation, Probe burden, deterministic rule, controlled comparison, and pilot readiness.

Version 6.3 | July 2026 | Three-option trigger and acceptance edition

1. Reviewer instructions

Complete ratings independently before discussion. Use the expert-review interface, start each scene from a fresh state, and judge the visible design as implemented. Proposed fixes belong in the comment fields and should not inflate the current rating. The profile screen is currently disabled and is not participant-facing functionality.

Rating	Meaning
1	Major problem / not ready
2	Substantial revision needed
3	Acceptable with minor revision
4	Strong / ready for cognitive testing

2. Method workflow

VPrivCal workflow: three-option trigger calibration

Expert-verified exposure labels combine with participant thresholds; reminder presentation stays fixed.



The study tests calibration under controlled expert-verified detections, not end-to-end VLM accuracy.

Review question	Rating / comment
Does the live workflow make clear that Q10 and Probe are two components of one method while the profile screen remains disabled?	1 2 3 4 Comment: _____
Is the 5-7 minute target credible when the required Hint walkthrough and category-complete review are included and timed separately?	1 2 3 4 Comment: _____
Is it clear that experts lock each cue as presence-only or sensitive-detail exposed before the participant policy runs?	1 2 3 4 Comment: _____

3. VPrivCal-Q10 candidate questions

Shared three-option trigger scale for Q1-Q6

1 Do not show reminders for this category | 2 Show reminders only when identifying or sensitive details are exposed | 3 Show reminders whenever this verified category is present

Item	Category / policy dimension	Relevance	Clarity	Actionability	Predictive value	Decision
Q1	Biometric data	1 2 3 4	1 2 3 4	1 2 3 4	1 2 3 4	Keep / Revise / Remove
Q2	Children images	1 2 3 4	1 2 3 4	1 2 3 4	1 2 3 4	Keep / Revise / Remove

Item	Category / policy dimension	Relevance	Clarity	Actionability	Predictive value	Decision
Q3	Personally identifiable information (PII)	1 2 3 4	1 2 3 4	1 2 3 4	1 2 3 4	Keep / Revise / Remove
Q4	Legal sensitivity information	1 2 3 4	1 2 3 4	1 2 3 4	1 2 3 4	Keep / Revise / Remove
Q5	Personal life	1 2 3 4	1 2 3 4	1 2 3 4	1 2 3 4	Keep / Revise / Remove
Q6	Background individuals	1 2 3 4	1 2 3 4	1 2 3 4	1 2 3 4	Keep / Revise / Remove
Q7	Inferred privacy risks	1 2 3 4	1 2 3 4	1 2 3 4	1 2 3 4	Keep / Revise / Remove
Q8	Other verified privacy categories	1 2 3 4	1 2 3 4	1 2 3 4	1 2 3 4	Keep / Revise / Remove
Q9	Uncertain detections	1 2 3 4	1 2 3 4	1 2 3 4	1 2 3 4	Keep / Revise / Remove
Q10	Task-irrelevant sensitive information	1 2 3 4	1 2 3 4	1 2 3 4	1 2 3 4	Keep / Revise / Remove

4. Item-specific prompts

Q1. Biometric data

Faces, fingerprints, and other distinctive body features that may identify a person.

Main concern / suggested wording: _____

Q2. Children images

Children, school events, playgrounds, and child-related activities.

Main concern / suggested wording: _____

Q3. Personally identifiable information (PII)

Financial or medical data; names, passports, email addresses, phone numbers, GPS, license plates, and other identifiers.

Main concern / suggested wording: _____

Q4. Legal sensitivity information

Nudity or explicit imagery; violent or unlawful acts; criminal acts; weapons or other high-risk items.

Main concern / suggested wording: _____

Q5. Personal life

Home interiors, personal items, location context, landmarks, routines, relationships, and activities.

Main concern / suggested wording: _____

Q6. Background individuals

Passersby, bystanders, coworkers, patients, patrons, and other people incidentally captured.

Main concern / suggested wording: _____

Q7. Inferred privacy risks

When a visual cue supports a sensitive inference, should the assistant show a brief privacy reminder?

Main concern / suggested wording: _____

Q8. Other verified privacy categories

If experts verify a privacy category not covered above, when should the assistant show a reminder?

Main concern / suggested wording: _____

Q9. Uncertain detections

When the assistant is unsure whether a cue is privacy-sensitive, should it show a brief privacy reminder?

Main concern / suggested wording: _____

Q10. Task-irrelevant sensitive information

When privacy-sensitive content is visible but not needed for the current task, should the assistant show a brief privacy reminder?

Main concern / suggested wording: _____

5. VPrivCal-Probe stimuli

VPrivCal-Probe: three egocentric contexts with category-complete review

Participants first point to any privacy-sensitive content they noticed. The interface then asks about every predefined category present in that scene.

Image 1 - Private: indoor family party



Available categories

Biometric Data Children Images PII

Legal Sensitivity Personal Life

Examples: family faces, child, school report card, wall-mounted rifle, home/family context.

Image 2 - Public: busy cafe



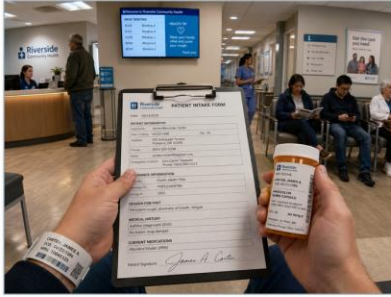
Available categories

Biometric Data Children Images PII

Background Individuals

Examples: patron faces, child, laptop/phone/receipt information, incidental bystanders.

Image 3 - Semi-public: hospital waiting area



Available categories

Biometric Data PII Background Individuals

Examples: patient/staff faces, intake form, medication label, wristband, waiting-room bystanders.

Overall coverage across the three scenes: Biometric Data, Children Images, PII, Legal Sensitivity Information, Personal Life, and Background Individuals.

Two questions per internal category-scene pair

A. Before the highlight: already aware / noticed but did not consider privacy / did not notice or realize VLM capability / not a concern.

B. Preferred trigger: never remind / remind only for identifying or sensitive-detail exposure / remind whenever the verified category is present.

Category identities remain hidden; participants see neutral linked-content labels.

The stimuli span private, public, and semi-public contexts. Coverage remains scene-level and internally category-complete across all six high-level categories. Phase A is unprompted pointing; Phase B uses neutral linked-content labels because participant-facing category identities are hidden.

Criterion	Scene 1	Scene 2	Scene 3	Comment
Context is realistic and distinct	1 2 3 4	1 2 3 4	1 2 3 4	_____
Internal category coverage is complete for the scene	1 2 3 4	1 2 3 4	1 2 3 4	_____
The optional evidence highlight is correctly localized and understandable	1 2 3 4	1 2 3 4	1 2 3 4	_____
The scene includes both obvious and subtle privacy content	1 2 3 4	1 2 3 4	1 2 3 4	_____
Point-first marking and neutral linked-content review are usable	1 2 3 4	1 2 3 4	1 2 3 4	_____
Cultural and ethical risk is acceptable	1 2 3 4	1 2 3 4	1 2 3 4	_____

Implemented sequence: require the five-step Hint before the first scene (practice data are not saved; Hint time is separate). Phase A shows a clean image: participants place/delete points or confirm none; category/detection matching is internal, unmatched points are retained, and only ambiguous overlaps trigger a neutral chooser. Phase B reviews every present internal category sequentially with neutral linked-content labels, an optional Show/Hide highlight, awareness status, and three-option reminder-trigger preference. Category identities remain hidden.

6. Probe-level review

Review question	Rating	Comment
Does the required five-step Hint teach point, delete, phase transition, response layout, and next-item navigation without contaminating real answers?	1 2 3 4	_____
Is it clear that Hint data are not saved and Hint time is excluded from Probe completion time?	1 2 3 4	_____
Does Phase A preserve an interpretable unprompted signal when categories and detections remain hidden and unmatched points are retained?	1 2 3 4	_____
Are neutral linked-content labels plus the Show/Hide highlight sufficient to identify the review target without category names?	1 2 3 4	_____
Are the awareness-status and three-option trigger items distinct, understandable, and sufficiently brief across every category?	1 2 3 4	_____
Are pointer/delete, overlap chooser, move/zoom/pan/reset, keyboard controls, and list fallback sufficiently usable and accessible?	1 2 3 4	_____

7. Held-out evaluation review

Held-out comparison under verified detections

All conditions receive the same locked cue labels and the same reminder presentation.

Generic policy

One fixed trigger threshold for every participant.

Q10-only policy

Three-option category priors plus binary cross-cutting preferences.

Full VPrivCal policy

Q10 priors corrected by category-by-context Probe responses.

Each cue still produces SHOW REMINDER or NO REMINDER.

Compare immediate acceptance and cue awareness without a presentation confound.

Metric / design decision	Expert judgment
Primary outcome: immediate acceptance rating (1-5) after each standardized reminder decision	Strong / Revise / Reject
Cue-awareness difference between generic and VPrivCal policies	Strong / Revise / Reject
Binary decision alignment, false-reminder rate, and missed-reminder rate	Strong / Revise / Reject

Metric / design decision	Expert judgment
Two-expert verification of category and exposure level before study inclusion	Strong / Revise / Reject
Balance presence-only and sensitive-detail-exposed held-out cues where feasible	Strong / Revise / Reject
Treat simulations as software checks and short-video outcomes as immediate only	Strong / Revise / Reject

7.1 Exposure-label codebook review

Experts first confirm category membership. Then ask whether the visible cue identifies or links a person, or exposes a private fact about a person or household. If yes, label SENSITIVE_DETAIL_EXPOSED; otherwise label PRESENCE_ONLY. Disagreements are adjudicated or excluded before the cue JSON is locked.

Category	PRESENCE_ONLY anchor	SENSITIVE_DETAIL_EXPOSED anchor
Biometric data	Distant, occluded, or otherwise non-recognizable person/body feature.	A clear face or distinctive body feature that supports recognition or re-identification.
Children images	A child is present without a recognizable identity or linkable school/location detail.	A recognizable child, readable name, school affiliation, home link, or other identity/location detail.
PII	A document, screen, badge, plate, or device is present but identifying content is unreadable.	A readable name, address, account/contact number, email, plate, patient ID, or comparable identifier.
Legal sensitivity	A potentially sensitive object or setting is present without person-specific sensitive content.	Explicit content, a visible unlawful/high-risk act, or person-linked legal/safety information.
Personal life	A generic home, routine, relationship, or personal object is visible without a linkable private fact.	An intimate activity, precise location, relationship/routine, or household detail is exposed and linkable.
Background individuals	An incidental person is distant or non-recognizable.	A bystander is recognizable or linked to a readable name, workplace, health context, or other private detail.

Reviewer judgment: Can two independent experts apply these anchors consistently? ☐ Strong ☐ Revise ☐ Reject
 Comment: _____

8. Final decision

VPrivCal-Q10: ☐ Proceed ☐ Minor revision ☐ Major revision ☐ Replace

VPrivCal-Probe interface (including Hint): ☐ Proceed ☐ Minor revision ☐ Major revision ☐ Replace

Workflow, expert verification, and immediate acceptance evaluation: ☐ Proceed ☐ Minor revision ☐ Major revision ☐ Replace

Simulated policy-output pre-verification: ☐ Proceed ☐ Minor revision ☐ Major revision ☐ Defer

Overall readiness for cognitive interviews: ☐ Yes ☐ Yes after revision ☐ No

Three highest-priority changes:

1. _____
2. _____
3. _____

Alternative idea or missing test: _____